Introductory Calculus

Section 1: Limits & Continuity

1 Evaluate $\lim_{x \to 2} \frac{x^3 - 8}{x - 2}$.

a. 12

b. 8

c. 4

d. Does not exist

2 Evaluate $\lim_{x \to 0} \frac{\tan(3x)}{x}$.

a. 3

b. 1

c. $\frac{1}{3}$

d. 0

3 What value of a makes $f(x) = \frac{x^2 - 1}{x - 1}$ for $x \neq 1$, and $f(1) = a$ continuous at $x = 1$?

a. 2

b. 1

c. 0

d. -1

Section 2: Derivatives & Power/Product Rules

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4 Find the derivative of $f(x) = 4x^3 - 5x^2 + 7x - 9$.

a. $12x^2 - 10x + 7$

b. $12x^2 - 5x + 7$

c. $4x^2 - 10x + 7$

d. $12x^3 - 10x^2 + 7$

5 What is the slope of the tangent line to $y = x^2e^x$ at $x = 1$?

a. $3e$

b. $2e$

c. e

d. $4e$

6 What is the derivative of $g(x) = \frac{2x+1}{x-3}$?

a. $-\frac{7}{(x-3)^2}$

b. $\frac{7}{(x-3)^2}$

c. $\frac{2}{(x-3)^2}$

d. $-\frac{5}{(x-3)^2}$

Section 3: Chain Rule & Extrema

7 Find the derivative of $h(x) = (3x^2 - 5)^4$.

a. $24x(3x^2 - 5)^3$

b. $4(3x^2 - 5)^3$

c. $12x(3x^2 - 5)^3$

d. $24(3x^2 - 5)^3$

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8 At what value of x does $f(x) = 2x^3 - 9x^2 + 12x + 1$ have a local minimum?

a. $x = 2$

b. $x = 1$

c. $x = 0$

d. $x = 3$

9 What is the acceleration of a particle at $t = 2$ if its position is $s(t) = t^3 - 6t^2 + 5t$?

a. 0

b. 5

c. -7

d. 6

Section 4: Definite Integrals & Fundamental Theorem

10 Evaluate the definite integral: $\int_0^3 (3x^2 - 2x + 1)dx$.

a. 21

b. 18

c. 24

d. 27

11 What is the area of the region bounded by $y = 4 - x^2$ and the x-axis?

a. $\frac{32}{3}$

b. $\frac{16}{3}$

c. 16

d. 8

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12 If $\int_1^4 f(x)dx = 7$, what is $\int_1^4 [2f(x) + 3]dx$?

a. 23

b. 17

c. 20

d. 26

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